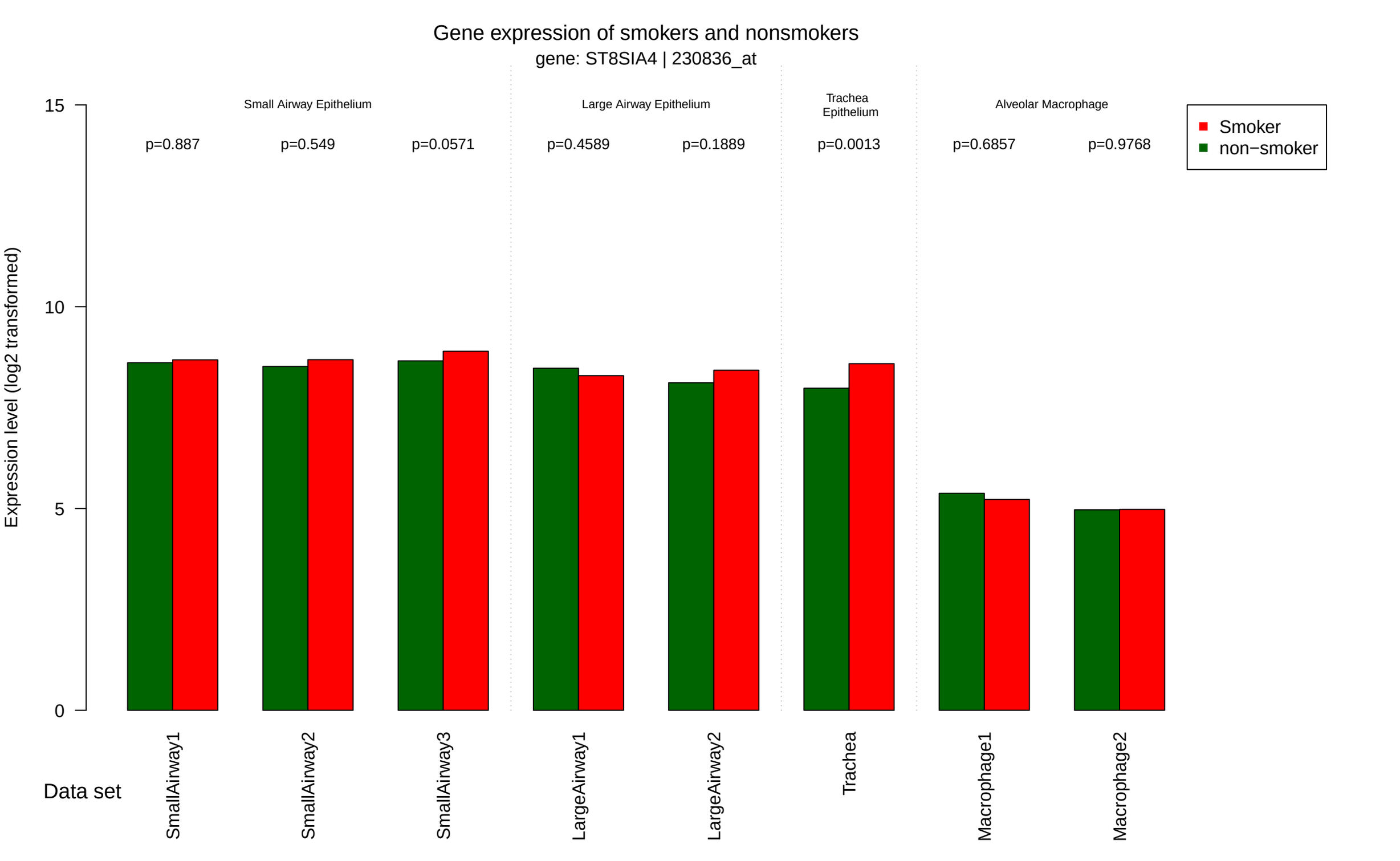
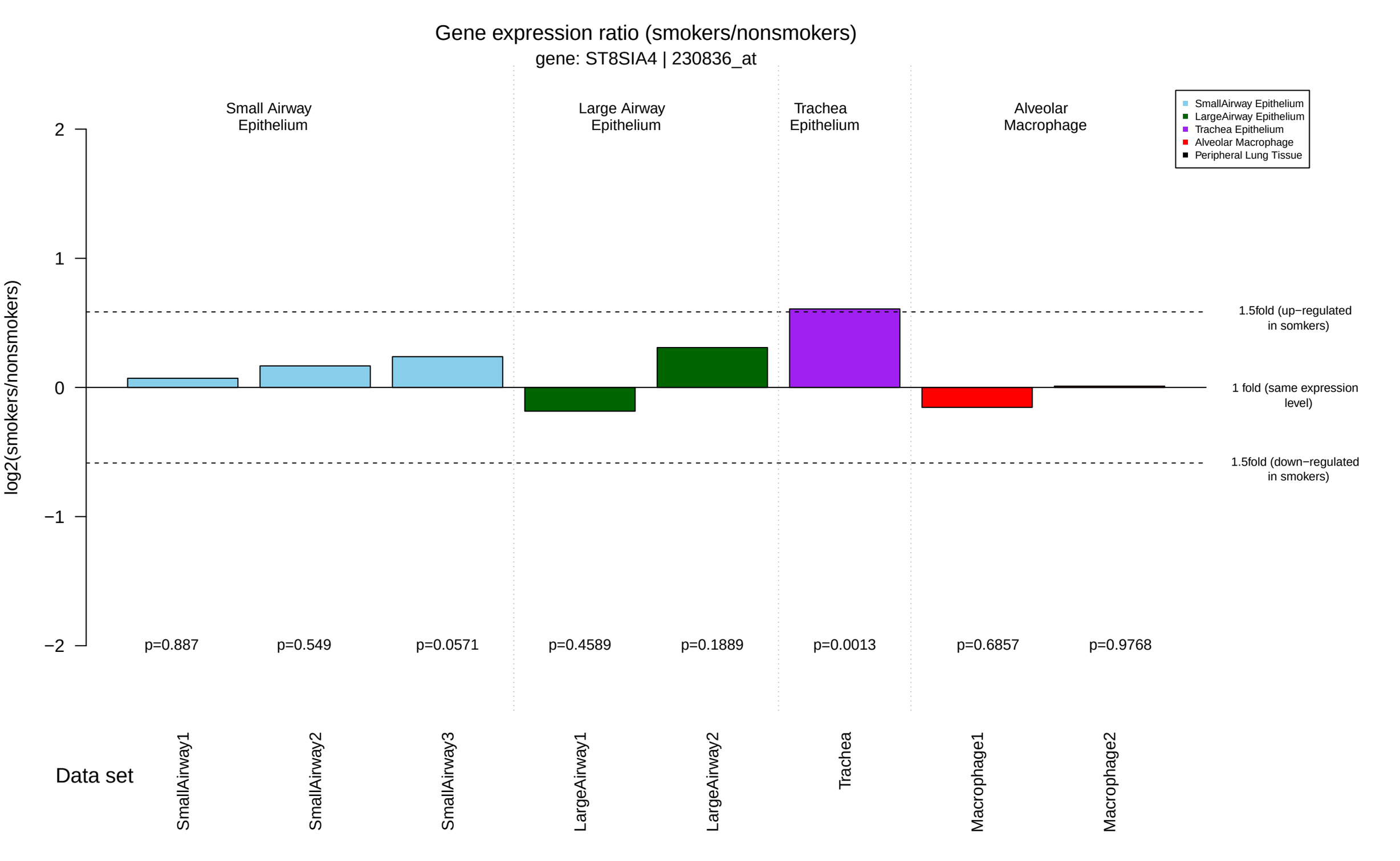
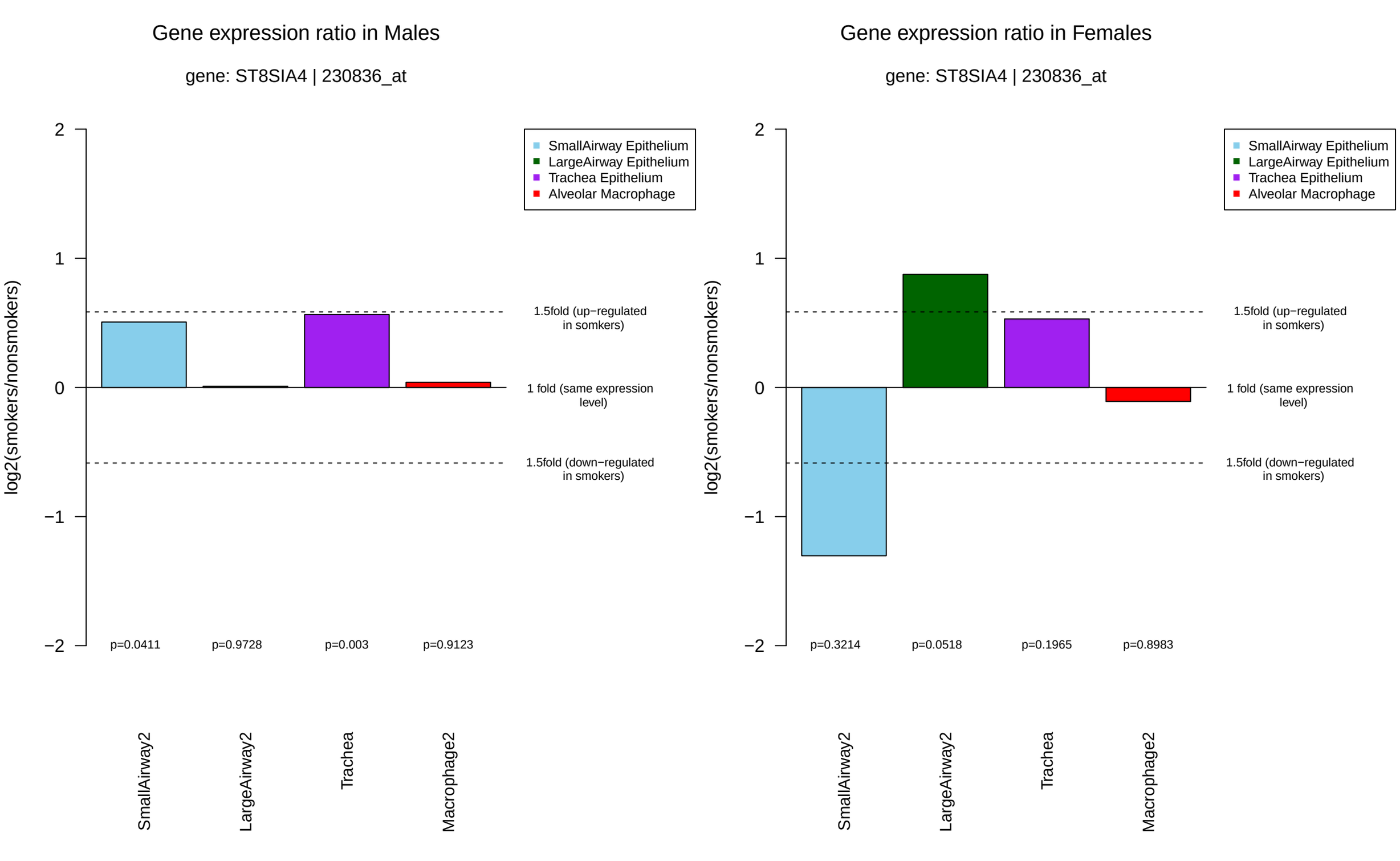
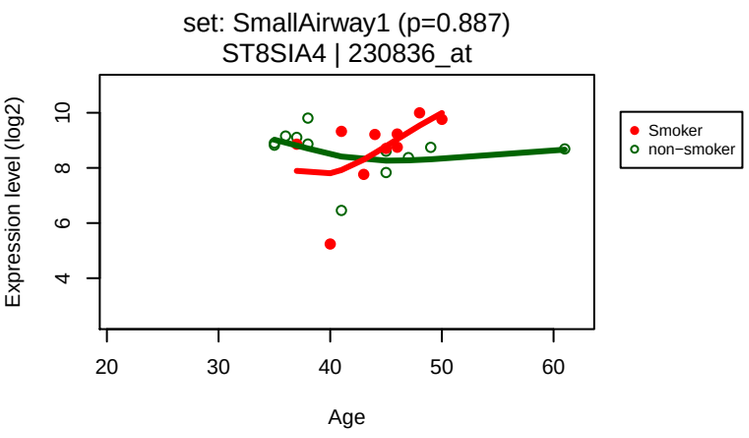
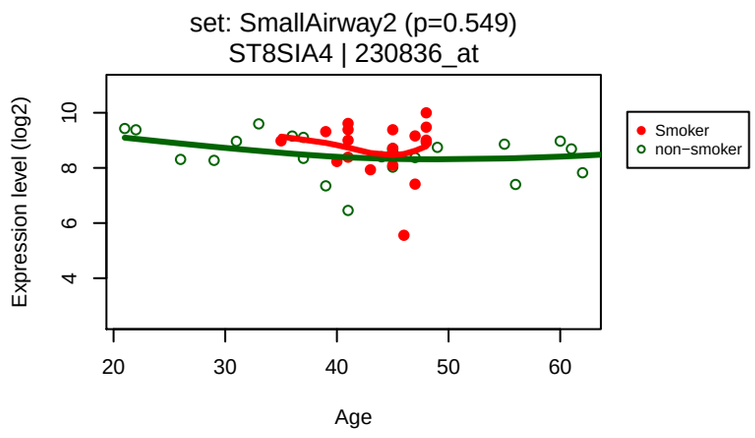
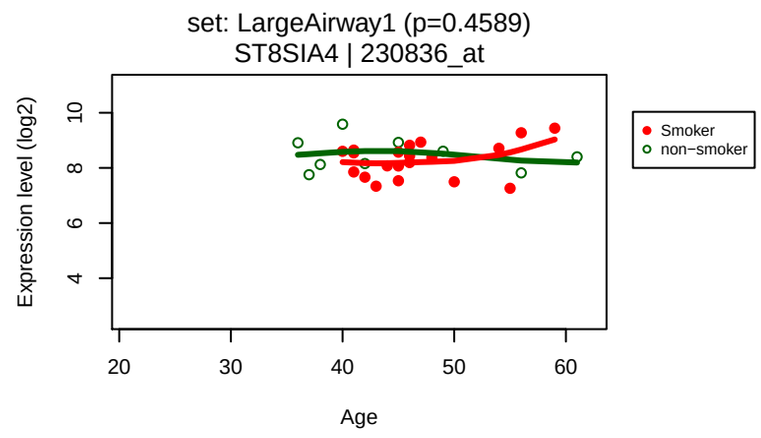

Gene expression of smokers and nonsmokers
gene: ST8SIA4 | 230836_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.887
p=0.549
p=0.0571
p=0.4589
p=0.1889
p=0.0013
p=0.6857
p=0.9768
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

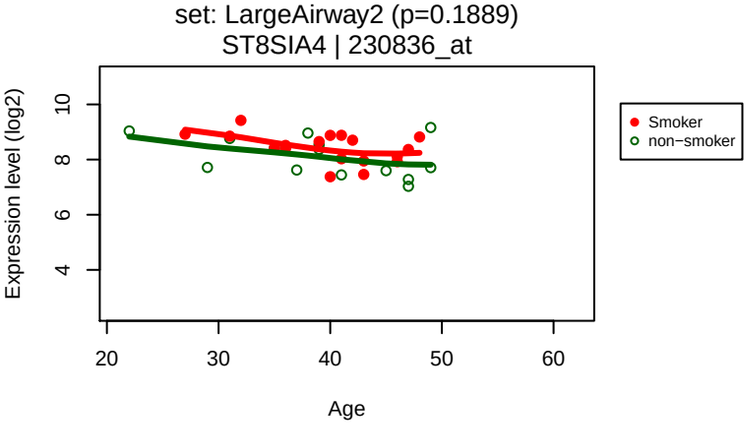

Gene expression ratio (smokers/nonsmokers)
gene: ST8SIA4 | 230836_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.887
p=0.549
p=0.0571
p=0.4589
p=0.1889
p=0.0013
p=0.6857
p=0.9768
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

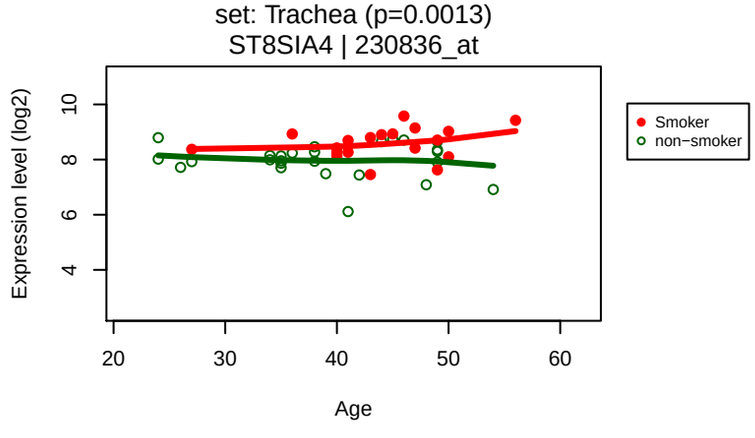

Gene expression ratio in Males
gene: ST8SIA4 | 230836_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0411
p=0.9728
p=0.003
p=0.9123
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: ST8SIA4 | 230836_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3214
p=0.0518
p=0.1965
p=0.8983
SmallAirway2
LargeAirway2
Trachea
Macrophage2

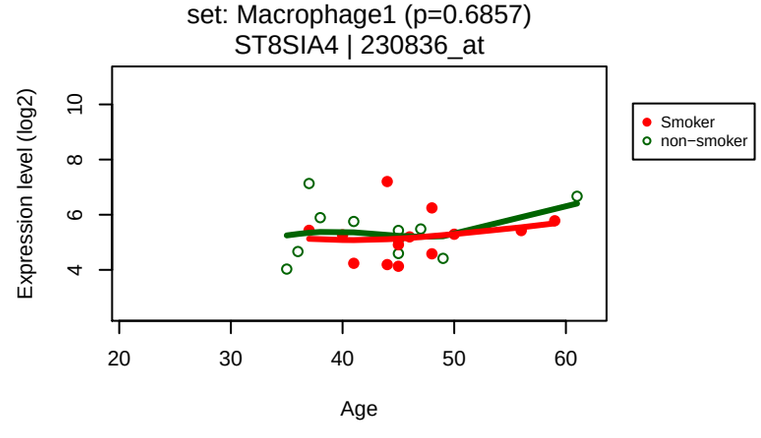

set: SmallAirway1 (p=0.887)
ST8SIA4 | 230836_at
Expression level (log2)
Age
Smoker
non−smoker

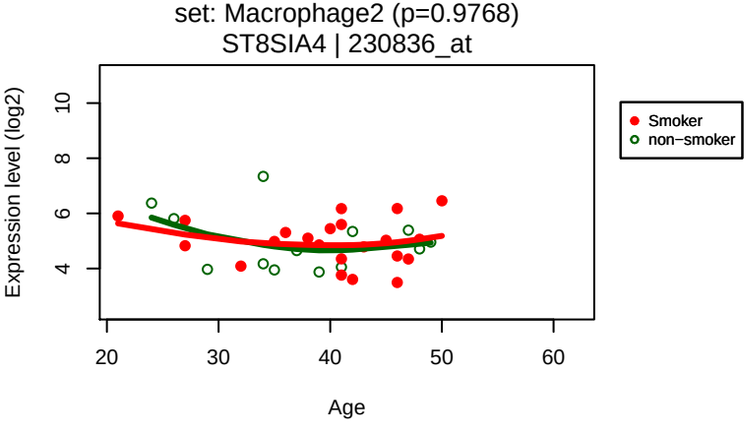

set: SmallAirway2 (p=0.549)
ST8SIA4 | 230836_at
Expression level (log2)
Age
Smoker
non−smoker

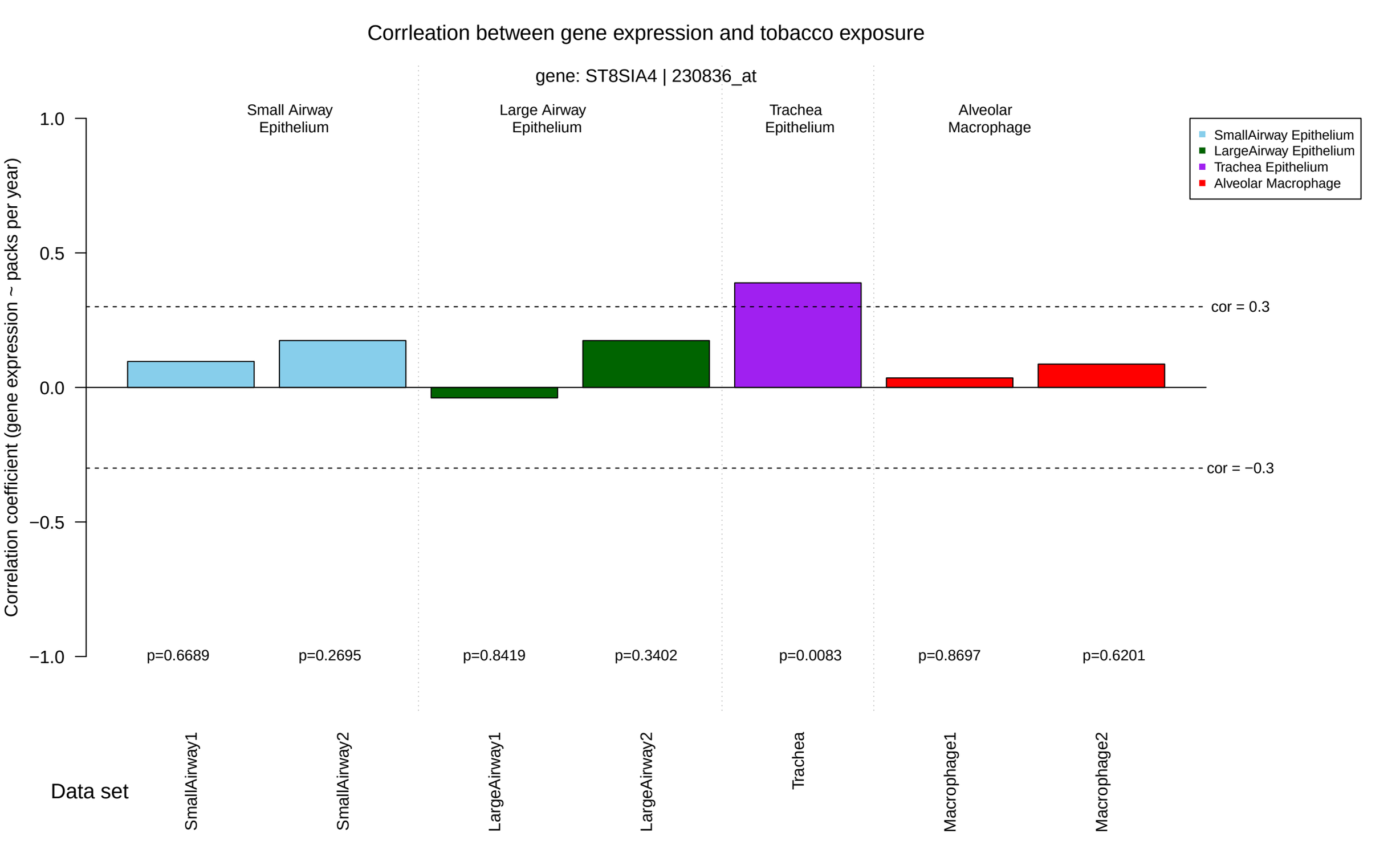

set: LargeAirway1 (p=0.4589)
ST8SIA4 | 230836_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.1889)
ST8SIA4 | 230836_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0013)
ST8SIA4 | 230836_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.6857)
ST8SIA4 | 230836_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.9768)
ST8SIA4 | 230836_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: ST8SIA4 | 230836_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.6689
p=0.2695
p=0.8419
p=0.3402
p=0.0083
p=0.8697
p=0.6201
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2